



COR 2MEDANA CH1-L.1003/PZeM-BP/AE

Number of Pumps	:	2
Rated Power	:	1.5kW
Rated Current	:	3.3A
Supply rating	:	3 x400V 50Hz
Duty/Assist/Standby	:	1D+1A
Starting Method	:	Multi VFD + Manual bypass DOL
IP rating	:	IP54
Ambient Temp.	:	40 deg C
Article Number	:	XXXX
Project Name	:	MADAR FARMS

Additional Specification:

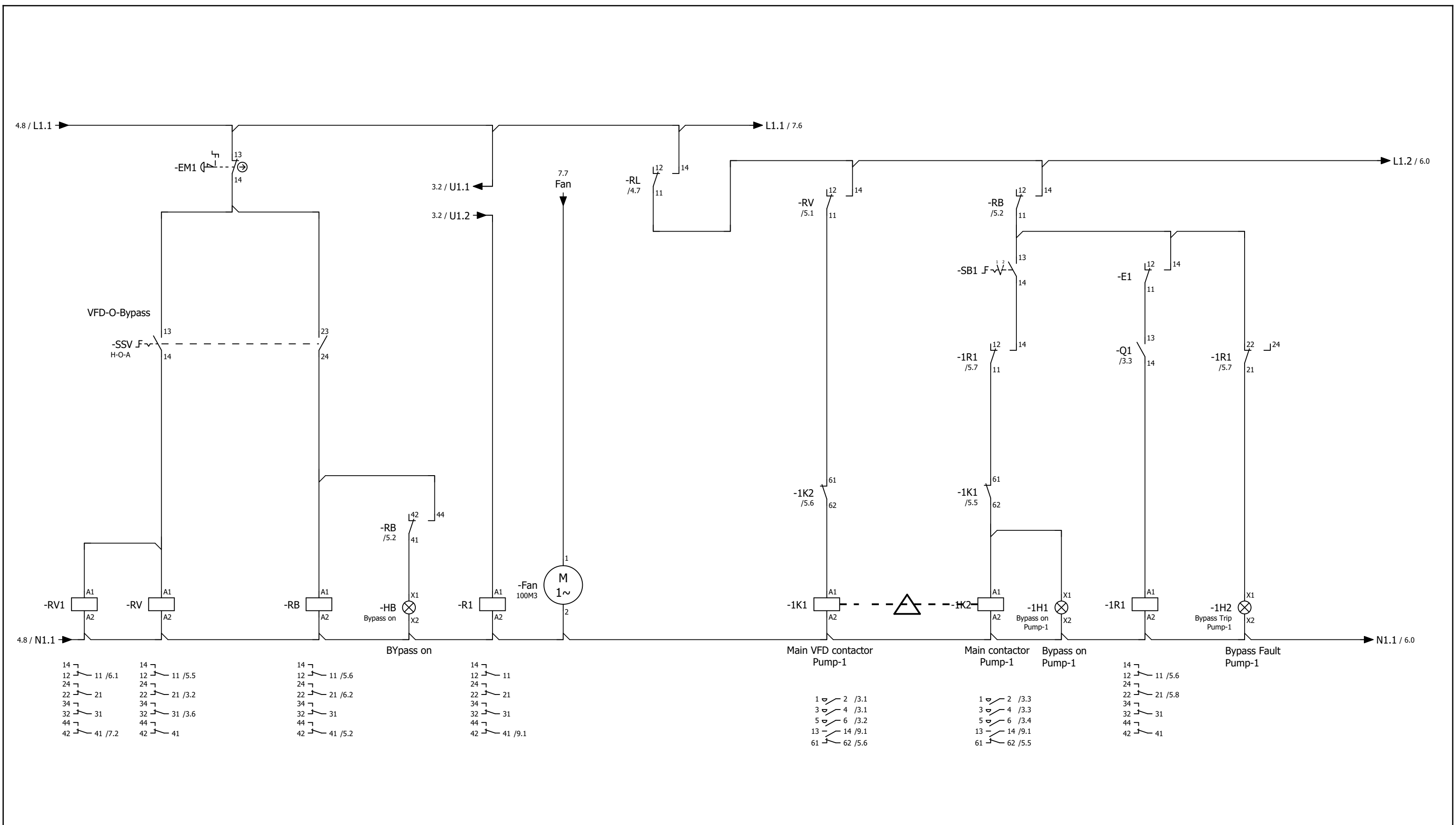
Manual bypass contactor
Bypass Run, Stop and trip indication

COR 2MEDANA CH1-L.1003-PZeM-BP-AE		A.N	 Wilo Middle East FZE Jebel Ali Free Zone–South PO Box 262720 Dubai UAE Tel: (+971 4) 880 91 77 Fax: (+971 4) 880 91 88. Ext. 122				
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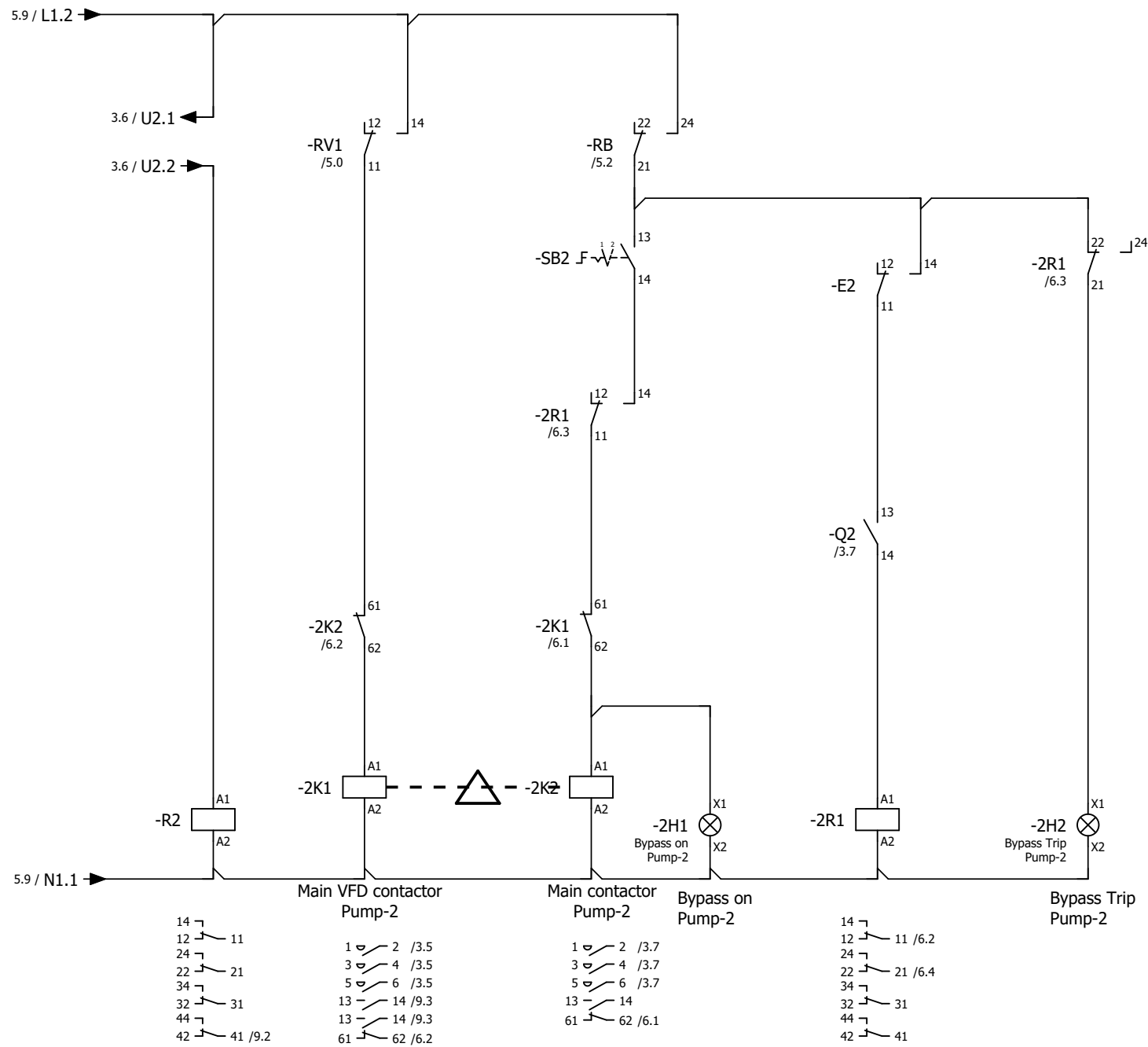
TECHNICAL DATA SHEET

<u>ENCLOSURE</u> SCHNEIDER ☐ ALFANAR ☐ Eurotech ☒ GRP ☐ <u>MATERIAL</u> SHEET STEEL ☒ STAINLESS STEEL ☐ ALUMINUM ☐ GRP ☐ <u>PRESENTATION</u> FIXED FRONT ☐ FUNCTIONAL DOOR ☒ PLAIN DOOR ☐ TRANSPARENT DOOR ☐ OTHER ☐ <u>TYPE OF CONSTRUCTION</u> FORM - 1 ☒ FORM - 3 ☐ FORM - 4 ☐ <u>DIMENSION (in mm)</u> HEIGHT : 1000 WIDTH : 600 DEPTH : 300 <u>COLOUR</u> RAL 7032 ☐ RAL 7035 ☒ <u>MOUNTING</u> WALL ☐ FLOOR ☒ PEDESTAL ☐ <u>INSTALLATION</u> INDOOR APPLICATION ☒ OUTDOOR APPLICATION ☐	<u>NEUTRAL/EARTH SYSTEM:</u> DIRECT TO EARTH (TT) ☐ SEPARATE (TN-S) ☒ EARTHED (TN-C) ☐ ISOLATED (IT) ☐ <u>BUS BARS</u> HARD DRAWN HIGH CONDUCTIVITY COPPER ELECTROLYTICALLY TINNED BUSBARS SUITABLY SIZED ☐ MAIN(VER.) Size(XXXX) ☐ MAIN(HOR.) Size(XXXX) ☐ NEUTRAL Size(XXXX) ☐ EARTH Size(XXXX) ☐ BRANCH Size(XXXX) ☐ INSULATION VOLTAGE UPTO 600V <u>CONNECTIONS</u> <table border="0"> <tr> <td>TERMINATION</td> <td>INC.</td> <td>OUTG.</td> </tr> <tr> <td>DIRECT TO DEVICE</td> <td>☒</td> <td>☐</td> </tr> <tr> <td>THRU TERMINAL BLOCKS</td> <td>☐</td> <td>☒</td> </tr> </table> <u>ACCESS</u> <table border="0"> <tr> <td>FRONT</td> <td>☒</td> <td>☒</td> </tr> <tr> <td>REAR</td> <td>☐</td> <td>☐</td> </tr> <tr> <td>LEFT SIDE/ RIGHT SIDE</td> <td>☐</td> <td>☐</td> </tr> </table> <u>CABLE ENTRY</u> <table border="0"> <tr> <td>TOP</td> <td>☐</td> <td>☐</td> </tr> <tr> <td>BOTTOM</td> <td>☒</td> <td>☒</td> </tr> </table> <u>SHROUDING</u> <table border="0"> <tr> <td>ALL LIVE PARTS</td> <td>☒</td> <td>☒</td> </tr> </table> <u>WIRING</u> <table border="0"> <tr> <td>CONTROL 1.0 Sqmm</td> <td>☒</td> </tr> <tr> <td>C.T WIRES 2.5Sqmm</td> <td>☐</td> </tr> </table>	TERMINATION	INC.	OUTG.	DIRECT TO DEVICE	☒	☐	THRU TERMINAL BLOCKS	☐	☒	FRONT	☒	☒	REAR	☐	☐	LEFT SIDE/ RIGHT SIDE	☐	☐	TOP	☐	☐	BOTTOM	☒	☒	ALL LIVE PARTS	☒	☒	CONTROL 1.0 Sqmm	☒	C.T WIRES 2.5Sqmm	☐	<u>AUXILIARY SUPPLY</u> <u>VOLTAGE</u> 240V AC ☐ 220V AC ☒ 110V AC ☐ 24V AC ☒ 24V DC ☐ <u>ORIGIN</u> EXTERNAL ☐ INTERNAL ☒ AUXILIARY TRANSFORMER ☒ AUXILIARY DC SOURCE ☐ <u>CIRCUIT MARKING</u> <u>WIRING ></u> Power Cable : Phase : Upto - 16sqmm - Red, yellow, Blue Above 16sqmm - black cable with color code sleeve Neutral : Black Earth : Yellow + Green Control Cable : 230V AC : Red+Black 110V AC : Orange 24VAC : Grey 12V/24V DC : Blue(+), White(-) Volt Free contact : Brown <u>LABELS</u> FACIA LEGEND PLATES OF 1.5mm PRINTED PVC WITH WHITE LETTERS ON BLACK BACKGROUND ALL LABELS WITH CHAMFERED EDGES DANGER LABELS WITH WHITE LETTERS ON RED BACKGROUND	Standard Specification: Door interlock Isolator Motor overload protection Pump run hour Equalizer Trip changeover Normally duty+assist function standby changeover will be done by operator manually Dry run interlock System enable/disable Low/high pressure cutoff Door components: Wilo Controller Indication in display (Run-Green, Trip-Red, Auto-White, OFF/hand - Blue, Standby-Grey) HOA switch per pump BMS: VFC contact: Run, Trip, Auto per pump Soft link: Modbus serial communication <u>TYPE OF DRAWING</u> ISSUED FOR APPROVAL ☒ ISSUED FOR PRODUCTION ☐ AS BUILT DRAWING ☐ ALL DIMENSIONS ARE IN mm
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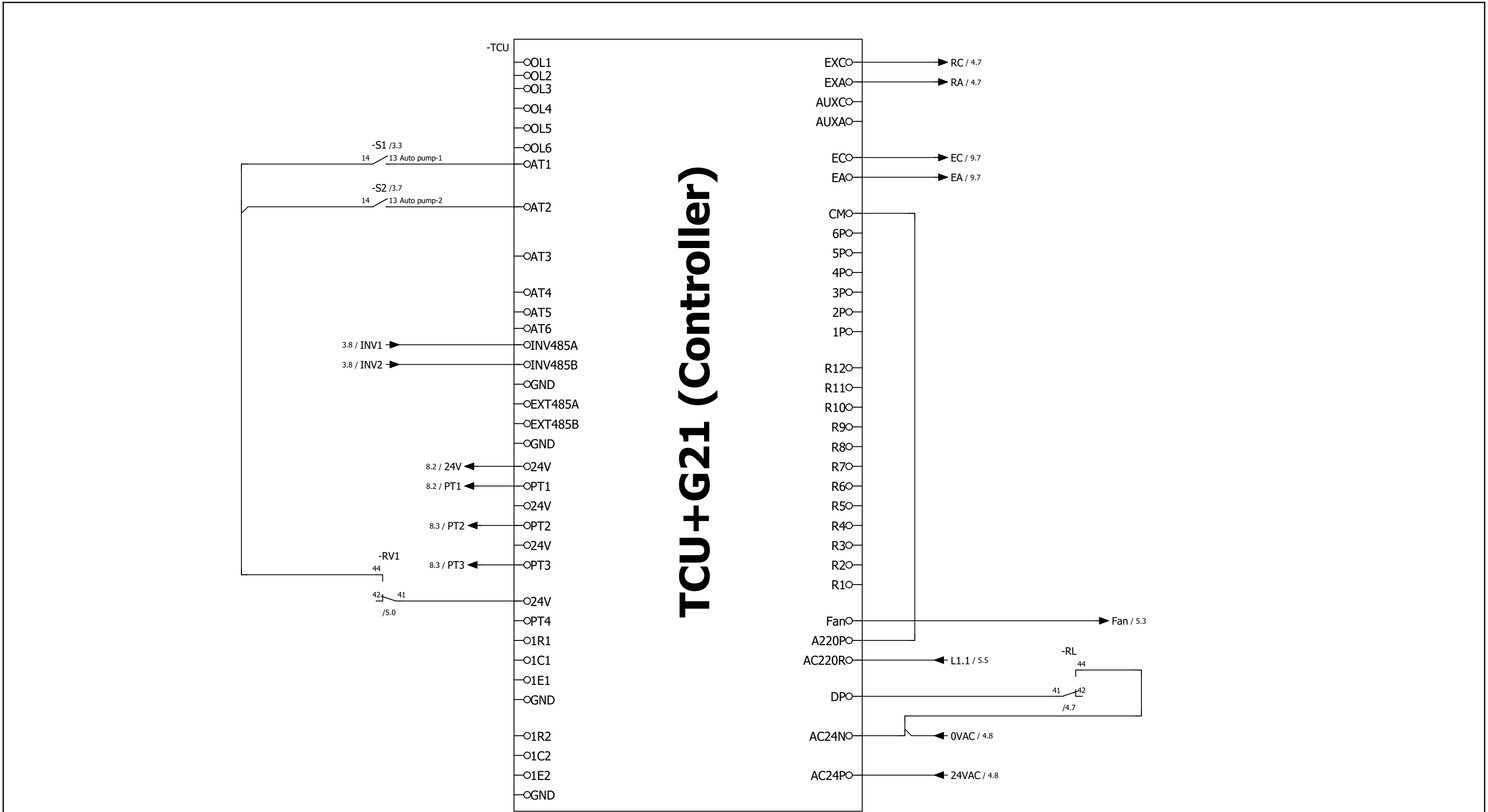
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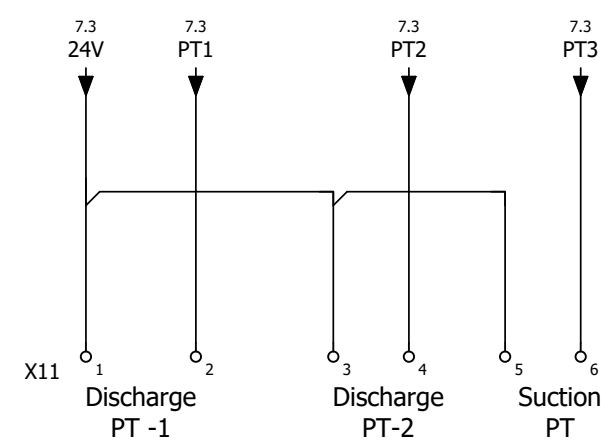
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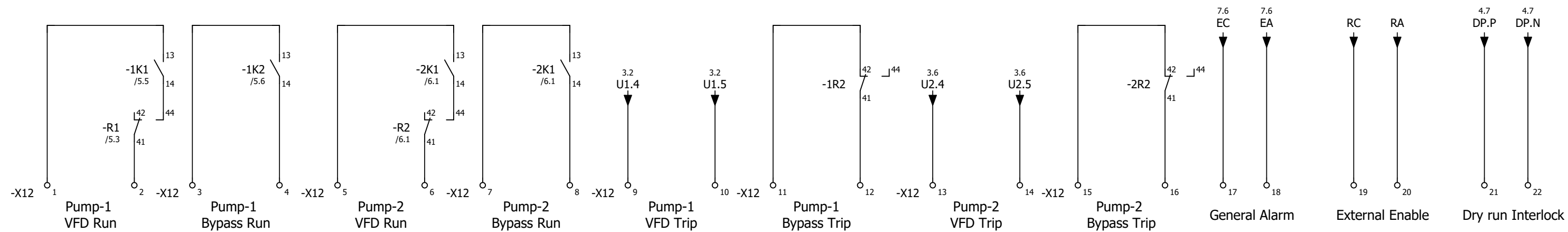


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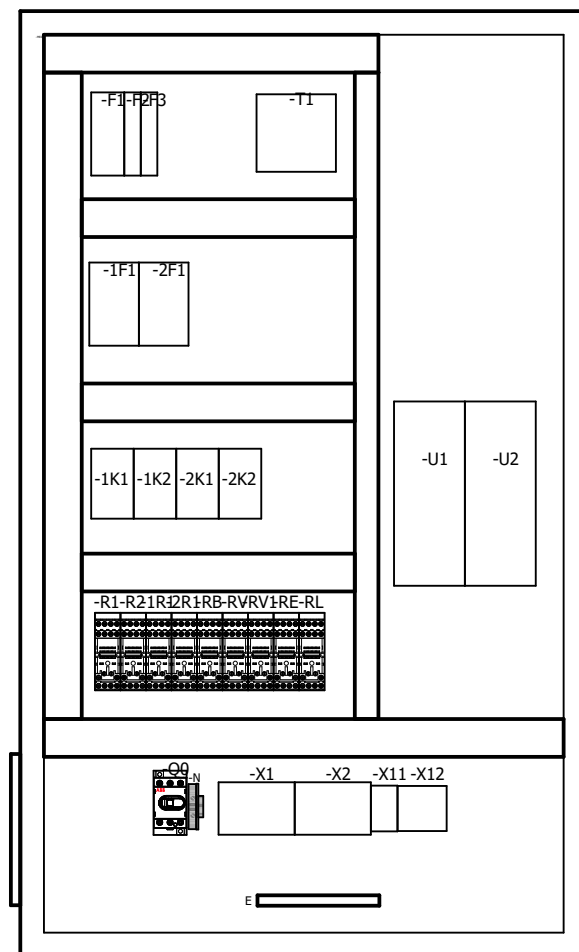
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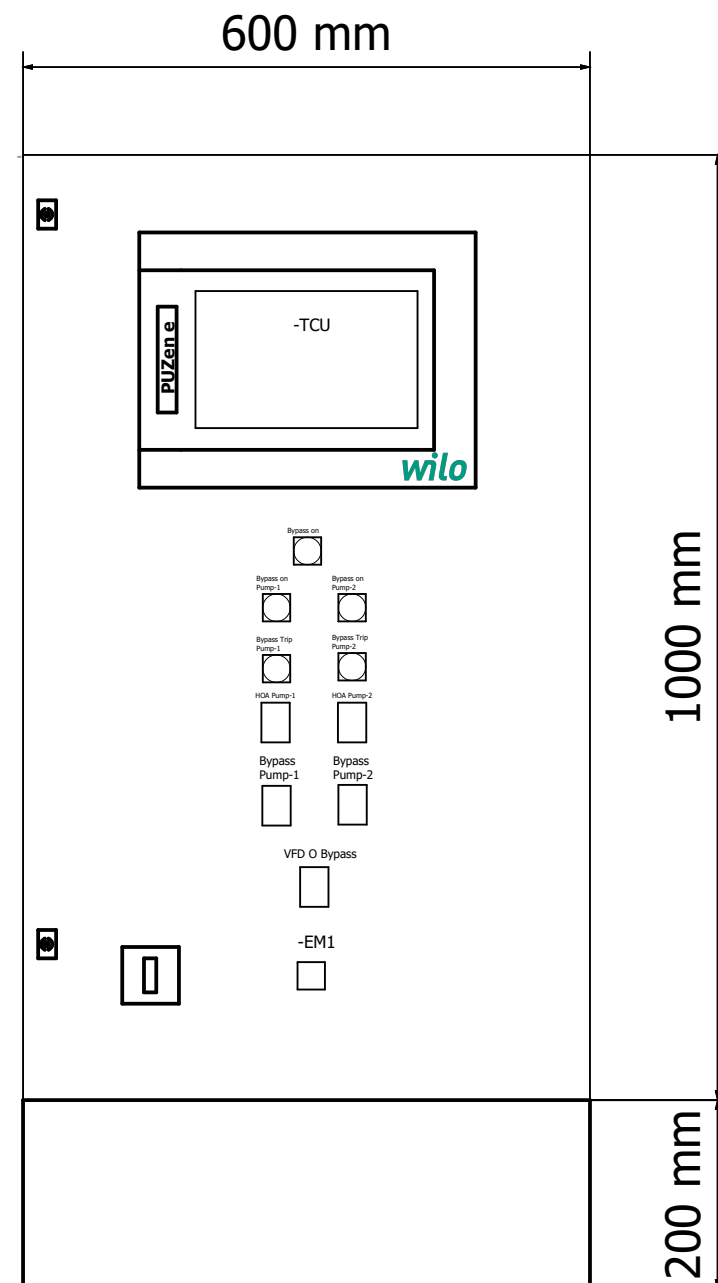
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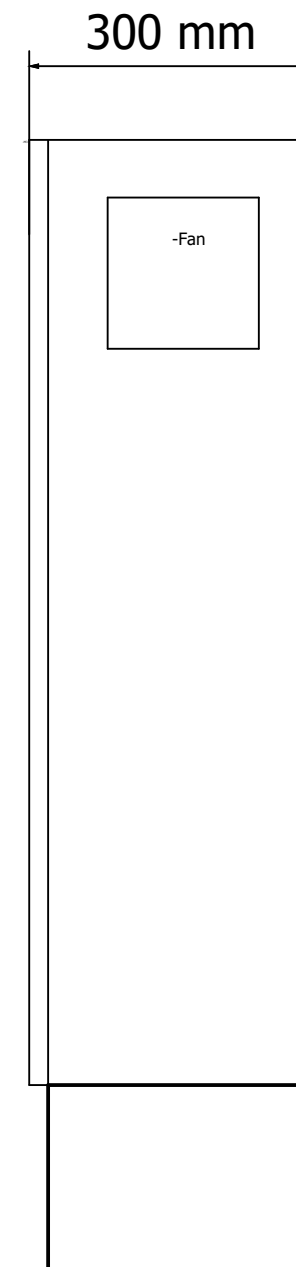
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Internal View



Front Fascia



Side view

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Summarized parts list

Device Tag	Quantity	Designation	Type number	Order number	Supplier
-EM1	1	EMERGENCY STOP PUSH BUTTON RED 1NO+1NC NC red		HLAY5ES545	Himel
-F1	1	Miniature Circuit Breaker - S202- 2P - C - 6 A 10kA	S202M-C6	2CDS272001R0064	ABB
-F2	1	CIRCUIT BREAKER 230/400V 10KA, 1-POLE, C, 6A	S201M	2CDS271001R0064	ABB
-F3	1	CIRCUIT BREAKER 230/400V 10KA, 1-POLE, C, 6A	S201M	2CDS271001R0064	ABB
-1F1;-2F1	2	CIRCUIT BREAKER 400V 10KA, 3-POLE, C, 10A	S203M	2CDS273001R0104	ABB
-Fan	1	PT PANEL FAN WITH FILTER 160X160	PTF	PTF1500	Plastim
-Fan	1	PT FILTER FOR FAN 160X160	PTI	PTI1500	Plastim
-1H1;-2H1	2	Monolithic pilot light Ø 22 - Green - integral LED - 230V		HLD1122C41N3	himel
-1H2;-2H2	2	Monolithic pilot light Ø 22 - Red - integral LED - 230V		HLD1122C41N4	himel
-HB	1	Monolithic pilot light Ø 22 - yellow - integral LED - 230V		HLD1122C41N5	himel
-1K1;-1K2;-2K1;-2K2	4	Contactor 12A 3P 230-240V AC 1NO auxilary	AX12-30-10-88	1SBL911074R8810	ABB
-M2	1	Metal Enclosure 1000H x 600W x300D IP55 RAL7035	ER	ER1006030D	
-N	1	Feed-through terminal block	TB 6 EI	3246036	PXC
-N	1	End-cover for 4 to 10	D-TB4/10 E	3246104	PXC
-N;-X1;-X2	5	End Bracket	E/TB	3246966	PXC
-Q0	1	OT40F3 switch-disconnector	OT40F3	1SCA104902R1001	ABB
-Q0	1	OHBS2AJ1 SELECTOR HANDLE	OHBS2AJ1	1SCA105215R1001	ABB
-Q0	1	OXS6X180 SHAFT for selector handle	OXS6X180	1SCA101659R1001	ABB
-Q1;-Q2	2	MS116-4.0 Manual Motor Starter	MS116-4.0	1SAM250000R1008	ABB
-Q1;-Q2	2	Aux.-contact for front mounting	HKF1-11	1SAM201901R1001	ABB
-1R1;-2R1;-R1;-R2;-RB;-RV;-RV1	7	CR-M230AC4 Pluggable interface relay 4c/o, A1-A2=230VAC, 250V/6A	CR-M230AC4	1SVR405613R3000	ABB
-1R1;-2R1;-R1;-R2;-RB;-RE;-RL;-RV;-RV1	9	CR-M4SS Standard socket for 2c/o or 4c/o CR-M relay	CR-M4SS	1SVR405651R3000	ABB
-RE;-RL	2	CR-M024AC4 Pluggable interface relay 4c/o, A1-A2=24VAC, 250V/6A	CR-M024AC4	1SVR405613R0000	ABB
-S1;-S2;-SSV	3	SELECTOR SWITCH 230VAC	HLAY5ED33	HLAY5ED33	Himel

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Summarized parts list

Device Tag	Quantity	Designation	Type number	Order number	Supplier
-SB1;-SB2	2	SELECTOR SWITCH 230VAC 2pos	HLAY5ED21	HLAY5ED21	Himel
-T1	1	Transformer 400/230V AC/24VAC 200VA	IT-100VA	IT-200-230V	Intact
-TCU	1	Controller Wilo for unidrive & multidrive	PCU+G22	3058933	Wilo
-U1;-U2	2	Variable frequency drive 1.5kW IP20 Bultin RFI, Bultin DC Choke,W/out Keypad, Modbus, BACnet	FC-101P1K5	131L9863	Danfoss
-X1;-X2	6	Feed-through terminal block 16sqmm	TB 16 EI	5792172	PXC
-X12;X11	28	Feed-through terminal block	TB 2,5 EI	3246010	PXC
	2	End-cover for 2.5	D-TB 2,5 E	3246094	PXC

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Mechnical Assembly

Pump Type : MEDANA CH1-L.1003-1/E/E/10T

Pump Article number : 4231497

Number of Pumps : 2

Rated power : 1.5kW

System Voltage : 3 x 400V AC 50Hz

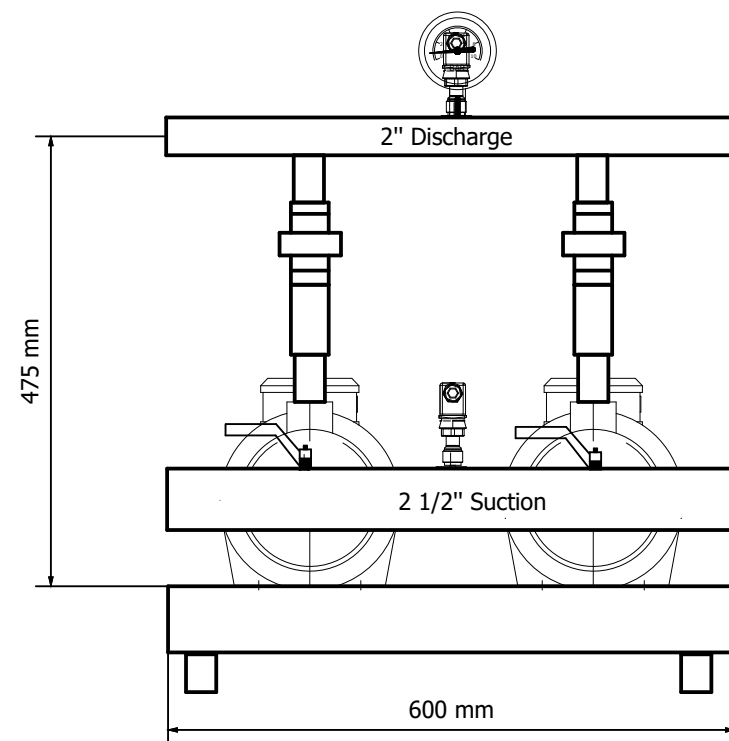
System Max. pressure : 16 bar

Range : Standard

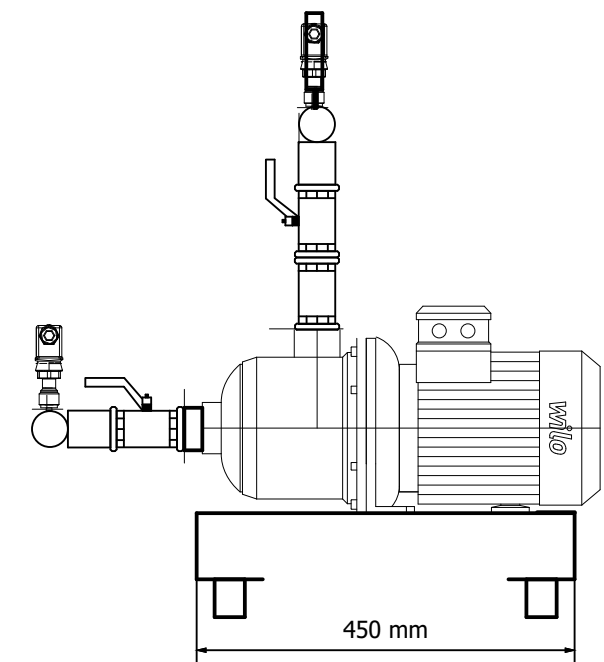
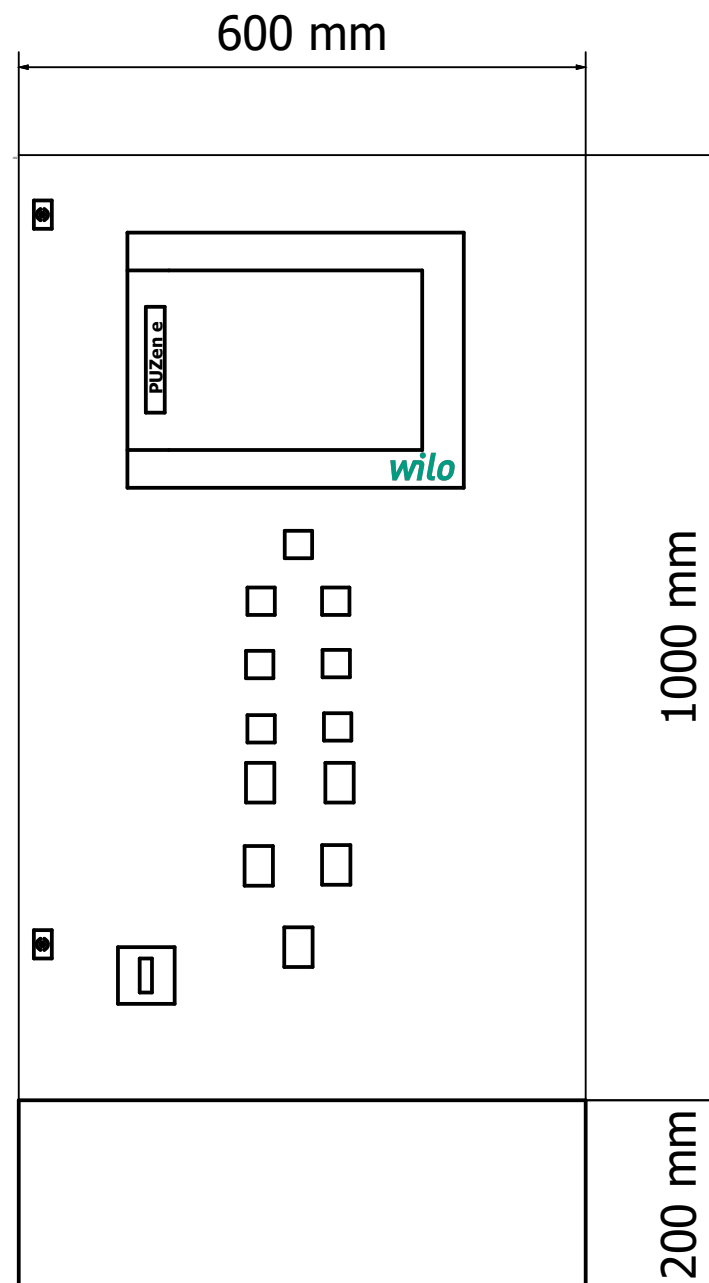
Speed : Variable speed

Adder:

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Front View



Side View

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Mechanical Assembly

Description	Type number	Qty	Supplier
Suction header SS304 2 1/2" 2 x 1 1/2" threaded PN16 300mm	A250/150	1	Wilo
Delivery header SS304 2" 2 x 1 1/4" threaded PN16 300mm	M200/125	1	Wilo
Check Valve 1 1/4"	-	3	Enolgas
Ball Valve 1 1/2"	-	3	Enolgas
Ball Valve 1 1/4"	-	3	Enolgas
Pressure Gauge 0 -10Bar		1	Wikai/Wilo
PRESS.SENSOR 1/4"0-16BAR 4-20MA M12 T528	2541620	1	Wilo
BASE-FRAME-EB-3*2/4/8/16H-2*2/4V-E300	4066174	1	Wilo
VIBRATION DAMPER 50x30mm,M10x28 Form D	502555494	6	Wilo

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